Inference under Gauss-Markov framework

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LSM

Distribution of Quadratic form of normally distributed random vectors

$X \sim N(\mu, \sigma^2)$

$$Z = \frac{X-\mu}{\sigma} \sim N(0,1)$$

MGF of Z

$$M_Z(t) = \mathbb{E}e^{tZ} = e^{t^2/2} \quad \forall t$$

Fact: Moment generating functions characterize the dist^{n} of a random variable if finite in a neighbourhood of 0.

$$M_X(t) = E(e^{tX}) = E(e^{t(\mu+\sigma Z)})$$
$$= e^{\mu t + \frac{1}{2}\sigma^2 t^2} \quad (\sigma = 0 \text{ allowed})$$

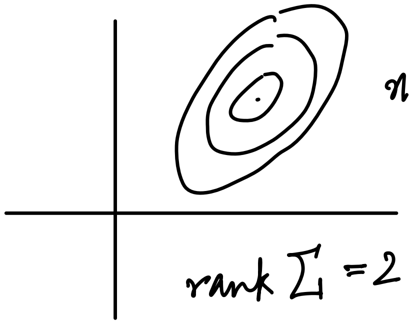
$$\left\{ \begin{array}{l} \underline{X}_{n\times 1} \sim N(\underline{\mu}_{n\times 1}, \Sigma_{n\times n}) \text{ where } \Sigma \text{ is an } n\times n \text{ positive semi-definite matrix} \\ \text{if and only if} \quad \underline{l}^T\underline{X} \sim N(\underline{l}^T\underline{\mu}, \underline{l}^T\Sigma\underline{l}) \ \forall l \end{array} \right.$$

Verification MGF of $\underline{X}$ is $M_{\underline{X}}(\underline{t}) = \mathbb{E}(e^{\underline{t}^T\underline{X}})$

$$M_{l^T\underline{X}}(\underbrace{s}_{\text{scalar}}) = \mathbb{E}(e^{s\underline{l}^T\underline{X}}) = M_{\underline{X}}(s\underline{l})$$

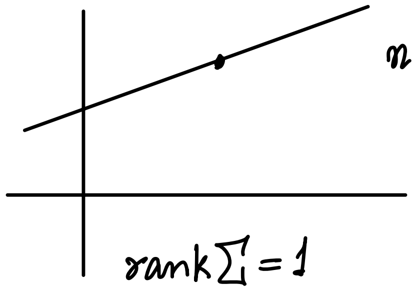
$$e^{sl^T\mu + \frac{s^2}{2}l^T\Sigma l}$$

<u>Remark</u>: If Σ is not positive definite then there exists l s.t $l^T\Sigma l = 0$. In that case, the distribution of X is singular that is it does not have a joint density.



$n=2$

rank $\Sigma = 2$



$n=2$

rank $\Sigma = 1$

$$\underset{\sim}{X} \sim N(\underset{n\times 1}{\underset{\sim}{\mu}}, \Sigma_{n\times n})$$

$$\Rightarrow C\underset{\sim}{X} \sim N(C\underset{\sim}{\mu}, C\Sigma C^T)$$

$$\underset{\sim}{X} = \begin{pmatrix} \underset{\sim}{X_1} \\ \underset{\sim}{X_2} \end{pmatrix} \sim N\left(\begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}, \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix}\right)$$

Σ is full rank iff Σ_{11} and $\Sigma_{22} - \Sigma_{21}\Sigma_{11}^{-1}\Sigma_{12}$ are non-singular.

Transformation (block diagonalization of Σ)

$$\begin{pmatrix} I & 0 \\ -\Sigma_{21}\Sigma_{11}^{-1} & I \end{pmatrix}\begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix}\begin{pmatrix} I & -\Sigma_{11}^{-1}\Sigma_{12} \\ 0 & I \end{pmatrix} = \begin{pmatrix} \Sigma_{11} & 0 \\ 0 & \Sigma_{22} - \Sigma_{21}\Sigma_{11}^{-1}\Sigma_{12} \end{pmatrix}$$

We can use this block diagonalization transformation to prove that
the conditional distribution of X_2 given X_1 is

$$N\left(\mu_2 + \Sigma_{21}\Sigma_{11}^{-1}(X_1 - \mu_1), \Sigma_{22} - \Sigma_{21}\Sigma_{11}^{-1}\Sigma_{12}\right)$$

Let $Y = L\begin{pmatrix} X_1-\mu_1 \\ X_2-\mu_2 \end{pmatrix} = \begin{pmatrix} X_1-\mu_1 \\ (X_2-\mu_2) - \Sigma_{21}\Sigma_{11}^{-1}(X_1-\mu_1) \end{pmatrix} = \begin{pmatrix} Y_1 \\ Y_2 \end{pmatrix}$

$$\sim N(0, L\Sigma L^T)$$

$$= N\left(0, \begin{pmatrix} \Sigma_{11} & 0 \\ 0 & \Sigma_{22} - \Sigma_{21}\Sigma_{11}^{-1}\Sigma_{12} \end{pmatrix}\right)$$

$Y_1 \perp\!\!\!\perp Y_2$ since $\mathrm{cov}(Y_1, Y_2) = 0$

$$Y_2 = X_2 - \mu_2 - \Sigma_{21}\Sigma_{11}^{-1}(X_1-\mu_1) \mid X_1 - \mu_1 = Y_1$$

$$\sim N(0, \Sigma_{22} - \Sigma_{21}\Sigma_{11}^{-1}\Sigma_{12})$$

$$\Rightarrow X_2 \mid X_1 \sim N(\mu_2 + \Sigma_{21}\Sigma_{11}^{-1}(X_1-\mu_1), \Sigma_{22} - \Sigma_{21}\Sigma_{11}^{-1}\Sigma_{12})$$

Remark: If Σ_{11} is singular, replace Σ_{11}^{-1} by Σ_{11}^{-} (any generalized inverse of Σ_{11})

Check that $\mathcal{C}(\Sigma_{12}) \subset \mathcal{C}(\Sigma_{11})$

and $(X_1-\mu_1) \in \mathcal{C}(\Sigma_{11})$ with prob-1

Show that $\Sigma_{12} b \in \mathcal{C}(\Sigma_{11})$ $\forall$ b

$$\Sigma_{12} b = \mathbb{E}(\overbrace{(X_1-\mu_1)}^{\Sigma_{11}W}(X_2-\mu_2)^T b)$$

$$= \Sigma_{11}\mathbb{E}(W(X_2-\mu_2)^T)b$$

$l^T(X_1-\mu_1)=0$ if $l^T\Sigma_{11}l=0$

$\Rightarrow X_1-\mu_1 \in$ the eigen subspace of Σ_{11} associated with the non-zero eigenvalues.

Observe that the statement

$$X_2|X_1 \sim N(\mu_2+\Sigma_{21}\Sigma_{11}^{-1}(X_1-\mu_1), \Sigma_{22}-\Sigma_{21}\Sigma_{11}^{-1}\Sigma_{12})$$

$$\Rightarrow \mathbb{E}(X_2|X_1)=\mu_2+\Sigma_{21}\Sigma_{11}^{-1}(X_1-\mu_1)$$

$\longrightarrow$ conditional expectation is linear in X_1

If $\dim X_2=1$ then let $\beta_0=\mu_2-\Sigma_{21}\Sigma_{11}^{-1}\mu_1$

$\beta_1=\Sigma_{11}^{-1}\Sigma_{12}$

$$X_2|X_1 \sim N(\beta_0+\beta^T X_1, \Sigma_{22}\ldots)$$

<u>Non-central χ^2</u>

Let $X_1,\ldots,X_n$ be independent with $X_i\sim N(\mu_i,1)$

Then $\sum_{i=1}^n X_i^2 \underbrace{}_{\text{Def}} \chi_n^2(\underbrace{\delta/2}_{\text{notation}})$ where $\delta=\sum_{i=1}^n \mu_i^2$ is the non-centrality parameter

Let P be an $n\times n$ orthogonal projection matrix (i.e. $P=P^T$ and $PP=P$)

then $X^TPX \sim \chi^2_{\text{rank}(P)}\left(\frac{\mu^TP\mu}{2}\right)$

If $\operatorname{rank} P = r \le n$ then $P = Q\Lambda Q^T$

$$\Lambda = \begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix} \qquad Q^TQ = QQ^T = I_n$$

$$X^TPX = Y^T\Lambda Y$$

where $Y = Q^TX \sim N\left(Q^T\mu, I_n\right)$

Suppose now that P_1 and P_2 are orthogonal projection matrix projecting onto mutually orthogonal subspaces

Then X^TP_1X and X^TP_2X are independent χ^2 random variables.

$$\|P_1X\|^2 \quad \|P_2X\|^2 \qquad \operatorname{Cov}(P_1X, P_2X)$$
$$= P_1P_2 = 0$$

Suppose $U \sim \chi^2_k(\delta/2)$

$V \sim \chi^2_r$

U, V independent

then $\dfrac{U/k}{V/r}$ is said to have a non-central $F_{k,r}(\delta/r)$ distribution